



**VIT<sup>®</sup>**  
**BHOPAL**

# **MULTI-DISEASE PREDICTION SYSTEM**

*A Machine Learning Based Risk & Symptom Prediction Application*

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**VITyarthi — Build Your Own Project**

Project Report / Presentation

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Course / Subject:-Fundamentals in AI and ML

GitHub Repository:-<https://github.com/amatulhasanzaidi/AI-ML-PROJECT>

# Problem Statement

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- Most people have no easy way to check their risk of conditions like diabetes or heart disease outside a clinical setting.
- Everyday illnesses (cold, flu, COVID-19, migraine, food poisoning) share overlapping symptoms and are often self-diagnosed incorrectly.
- Goal: build a single system that validates input, predicts risk/illness, stores prediction history, and reports on both the data and the models.

# Objectives & Target Users

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## Objectives

- Train and serve 3 independent classification models
- Validate every input before it reaches a model
- Persist every prediction (CRUD) for later review
- Turn history and model metrics into readable analytics

## Target Users

- Students / instructors evaluating applied ML and software engineering practice
- Hobbyist learners exploring a runnable ML-backed console application

# Functional Requirements

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The system is organised into three major functional modules:

## 1. Data Management

- Generate and clean datasets (diabetes, heart, symptoms)
- Train/test split with feature scaling

## 2. Prediction Engine

- Diabetes & Heart Disease risk (Random Forest)
- Symptom → Illness checker (Decision Tree)

## 3. Records & Reporting

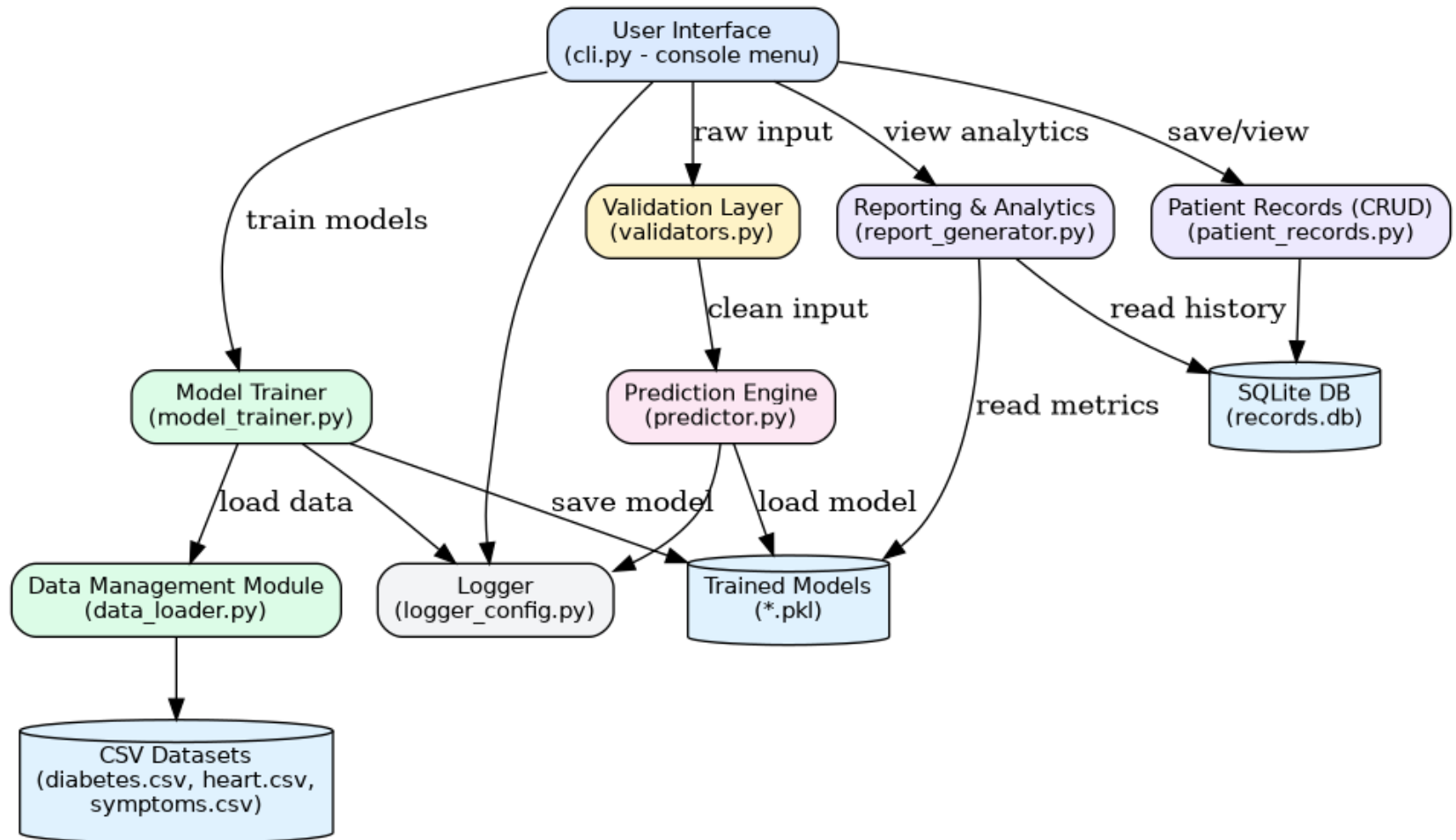
- CRUD patient history stored in SQLite
- Text summary + chart analytics

# Non-Functional Requirements

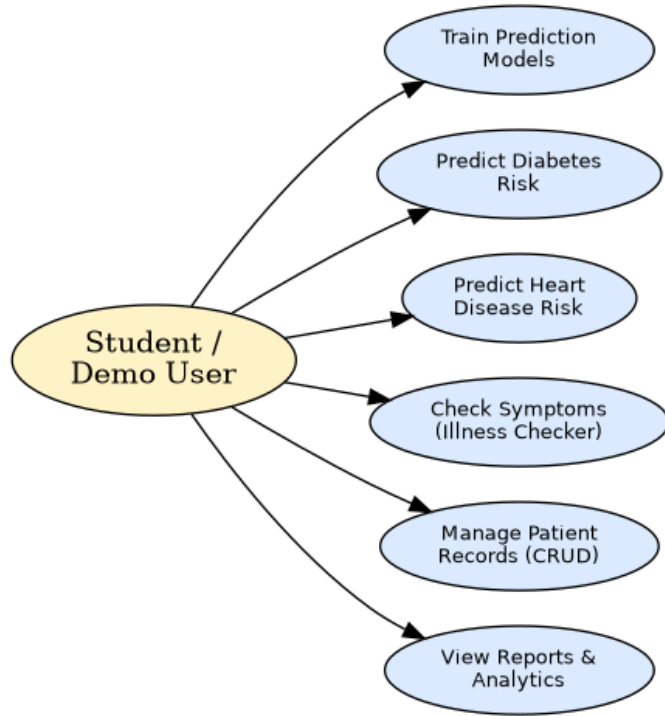
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|                                     |                                                                |
|-------------------------------------|----------------------------------------------------------------|
| <b>Performance</b>                  | Sub-second training; near-instant predictions                  |
| <b>Security</b>                     | No network calls; all data stored locally                      |
| <b>Reliability</b>                  | try/except around every action; app never crashes on bad input |
| <b>Scalability</b>                  | Independent modules; dataset/model can be swapped easily       |
| <b>Maintainability</b>              | Each module has a single, documented responsibility            |
| <b>Logging &amp; Error Handling</b> | Rotating file logger + custom ValidationError class            |

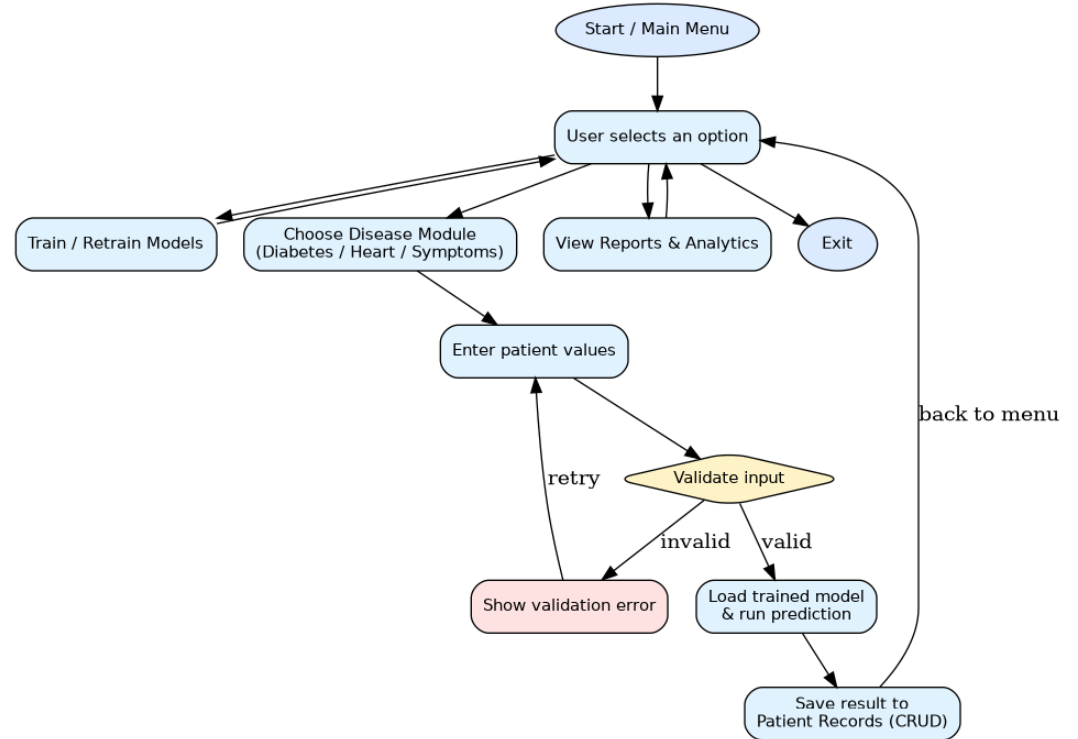
# System Architecture



# Use Case & Process Flow Diagrams



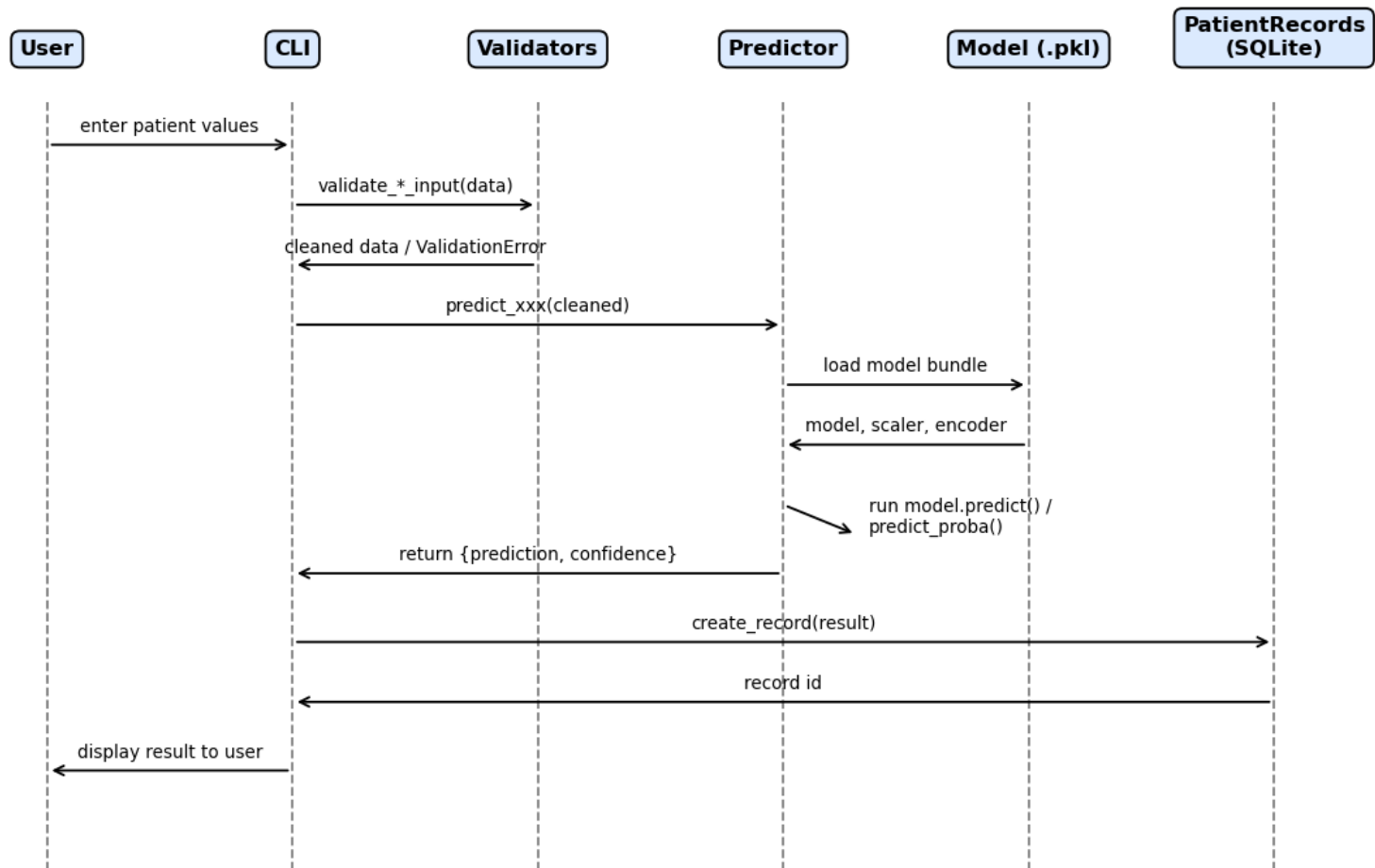
Use Case Diagram



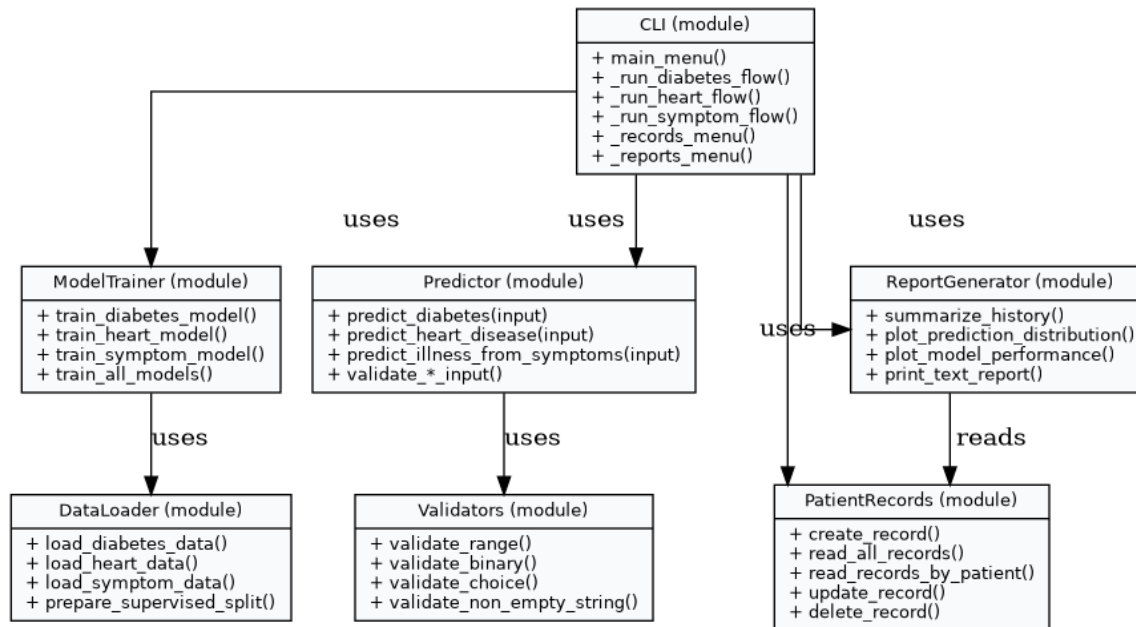
Process Flow Diagram

# Sequence Diagram

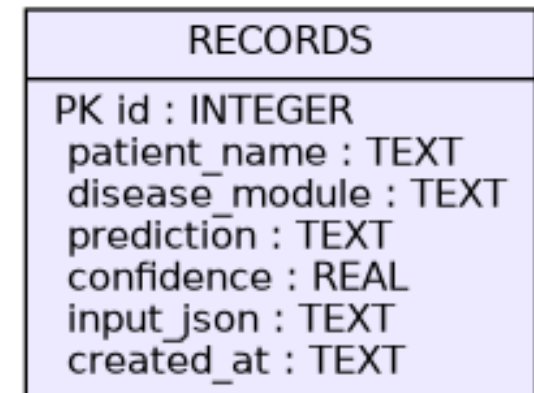
Sequence Diagram — Running a Disease Prediction



# Class Diagram & Database Design



Class / Component Diagram



ER Diagram

# Screenshots / Results

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MULTI-DISEASE PREDICTION SYSTEM (VITyarthi Project)
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Disclaimer: Educational demo only. Not medical advice.

=== MAIN MENU ===
1. Train / Retrain Prediction Models
2. Predict Diabetes Risk
3. Predict Heart Disease Risk
4. Symptom-Based Illness Checker
5. Patient Record Management (CRUD)
6. Reporting & Analytics
7. Exit
Choose an option: 2

--- Diabetes Risk Prediction ---
Patient name: Aarav Sharma
Pregnancies (0-20): 2
Glucose level mg/dL (40-260): 190
Blood pressure mmHg (20-140): 88
Skin thickness mm (0-100): 32
Insulin level mu U/mL (0-900): 250
BMI (10-70): 38.5
Diabetes pedigree function (0.0-3.0): 0.9
Age (1-120): 55

Result: Positive (confidence: 86.4%)
Saved as record #1.
```

| ID | Patient      | Module          | Prediction | Conf% | Created At          |
|----|--------------|-----------------|------------|-------|---------------------|
| 5  | Karan Mehta  | Symptom Checker | Migraine   | 97.5  | 2026-09-14T04:20:11 |
| 4  | Sneha Iyer   | Heart Disease   | Negative   | 92.75 | 2026-09-14T04:20:11 |
| 3  | Rohan Gupta  | Heart Disease   | Positive   | 85.58 | 2026-09-14T04:20:11 |
| 2  | Priya Nair   | Diabetes        | Negative   | 98.11 | 2026-09-14T04:20:11 |
| 1  | Aarav Sharma | Diabetes        | Positive   | 86.4  | 2026-09-14T04:20:11 |

```
=== PREDICTION HISTORY SUMMARY ===
Total predictions made : 5
```

By module:

- Symptom Checker: 1
- Heart Disease: 2
- Diabetes: 2

By outcome:

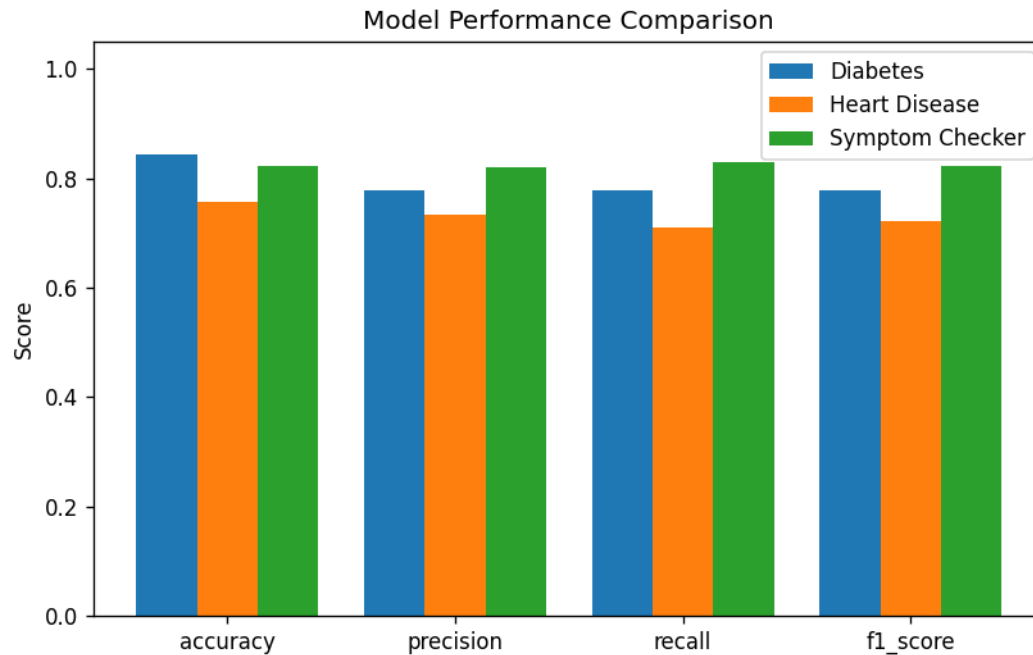
- Symptom Checker - Migraine: 1
- Heart Disease - Negative: 1
- Heart Disease - Positive: 1
- Diabetes - Negative: 1
- Diabetes - Positive: 1

```
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```

*Main Menu & Sample Prediction*

*Patient Records & Analytics*

# Model Performance



| Model              | Accuracy | F1    |
|--------------------|----------|-------|
| Diabetes (RF)      | 0.844    | 0.778 |
| Heart Disease (RF) | 0.757    | 0.721 |
| Symptom Checker    | 0.822    | 0.823 |

*Metrics measured on a held-out 20% test split.*

# Testing Approach

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- Validator unit tests — range, binary, choice, and non-empty-string checks
- Data loader tests — dataset columns and row counts
- End-to-end model tests — trains all 3 models; asserts  $\geq 60\%$  accuracy each
- Prediction tests — valid input accepted, out-of-range input rejected
- CRUD tests — isolated temp database: create, read, update, delete
- Result: 21 / 21 automated tests passing (pytest)